

DURGESH KUMAR ROUT

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience designing, training, and deploying production-grade AI applications. Proficient in Machine Learning, NLP, and full-stack Web Development. Delivered two live ML systems — a Fake News Detector and a Malicious URL Classifier. Strong foundation in Data Structures & Algorithms, OOP, and software engineering. Passionate about practical AI at the intersection of cybersecurity and intelligent systems.

TECHNICAL SKILLS

Languages	Python, Java, C, C++
ML & AI	Scikit-learn, Pandas, NumPy, NLP, TF-IDF Vectorization, Logistic Regression, Feature Engineering, Model Evaluation (Precision, Recall, F1)
Web Development	HTML5, CSS3, JavaScript, React.js, Streamlit
Tools	Git, GitHub, VS Code, Chrome DevTools, Hugging Face Spaces, Streamlit Cloud
CS Concepts	Data Structures & Algorithms, OOP, DBMS, Operating Systems, REST APIs

PROJECTS

Fake News Detection System — NLP & Machine Learning 2026
[Live Demo](#) — [GitHub Repository](#)

- Built an end-to-end NLP text classification pipeline to detect fake vs. real news articles using TF-IDF Vectorization and Logistic Regression, achieving strong accuracy on benchmark datasets.
- Applied robust preprocessing — tokenization, stopword removal, punctuation filtering — and managed the full ML lifecycle from data ingestion through model training, evaluation, and cloud deployment.
- Deployed an interactive Streamlit web application on Hugging Face Spaces enabling real-time article classification accessible publicly via URL.

Malicious URL Detection System — Cybersecurity ML 2026
[Live Demo](#) — [GitHub Repository](#)

- Developed a supervised ML classifier to identify phishing and malicious URLs by engineering lexical features: URL length, special character counts, digit ratios, subdomain depth, and TLD patterns.
- Benchmarked multiple classifiers on precision, recall, F1-score, and AUC-ROC; prioritised low false-negative rate for security-critical deployment.
- Built a real-time prediction interface and deepened expertise in cybersecurity-focused ML workflows, imbalanced dataset handling, and model interpretability.

FitHub — Health & Fitness Tracker Web Application 2025

- Architected a full-stack health and fitness web app with JWT-based authentication, BMI calculator, calorie tracker with macro breakdown, workout manager, and a structured diet planning module.
- Built a modular React.js frontend with responsive layouts and clean state management; applied DRY principles and Git-based version control for a maintainable, extensible codebase.

EDUCATION

B.Tech in Computer Science & Engineering 2023 – 2027

Institution of Technical Research and Engineering CGPA: 7.97 / 10.0

Class XII — Science Stream 2021 – 2023

Rotary Public School Percentage: 72.33%

Class X 2020 – 2021

Cement Nagar English Medium School Percentage: 84.8%

ACHIEVEMENTS & ACTIVITIES

- Deployed two production-grade ML applications with live public URLs on Streamlit Cloud and Hugging Face Spaces, demonstrating full end-to-end ownership.
- Participated in college-level hackathons; consistently practising DSA (arrays, trees, graphs, DP) for software engineering interview readiness.
- Interested in AI/ML applications, Cybersecurity, IoT and modern web technologies.
- Freelance video editor with real client experience; actively self-learning deep learning (TensorFlow/PyTorch) and advanced ML optimisation techniques.